T3000 API & Protocol

T3000 Center DB Configuration · built Nov 13, 2026

T3000 Center DB Configuration

Use this guide to configure a shared central SQL Server database in T3000. This allows multiple T3000 PCs on the same network to share trend logs and device data through one central Microsoft SQL Server instance.

Before starting, make sure SQL Server Express is installed and TCP/IP is enabled. See the SQL Server Express Setup guide if needed.

Step 1: Open T3000 and Update to the Latest Version

Open the T3000 Building Automation System desktop application.

Before configuring the shared database, make sure T3000 is up to date. Click **Help — Check for Updates** in the menu bar and install any available update. The Shared DB feature requires the latest version to work correctly.



Step 2: Go to the Dashboard

In the T3000 webview, click **Dashboard** in the top navigation bar. You can also open the Dashboard directly in a browser:

- **On this PC:** http://localhost:9103/#/t3000/dashboard
- **From another PC on the same network:** `http://<this-PC-IP>:9103/#/t3000/dashboard` (e.g. `http://192.168.1.22:9103/#/t3000/dashboard`)

You will see two mode cards:

- **Standalone** — stores data locally in SQLite on this PC (currently Active)
- **Shared DB** — connects to a central SQL Server database shared across multiple T3000 PCs

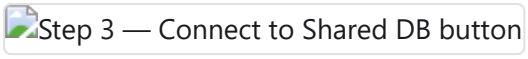
The **Shared DB** card shows *Not configured. Click to connect to a shared SQL Server database.* Click anywhere on the Shared DB card to begin setup.


Step 2 — Dashboard showing Standalone active and Shared DB not configured

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Step 3: Click "Connect to Shared DB"

After clicking the Shared DB card, a **Connect to Shared DB** button appears. Click it to open the Database Configuration page.


Step 3 — Connect to Shared DB button

Step 4: Database Configuration Page

The **Database Configuration** page opens, showing:

- **Centralized Trendlog & Data Storage** — description of the feature
- **Database Status** — currently shows *Standalone · Online*
- **Server / Client Configuration** — toggle is currently **Server Database: Disabled**


Step 4 — Database Configuration page, standalone mode, server database disabled

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Step 5: Enable Server Database and Select Role

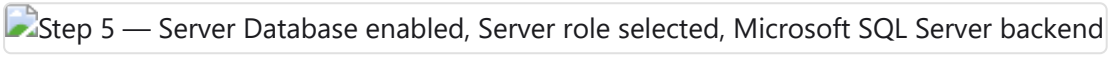
Toggle **Server Database** to **Enabled**.

Two role cards appear:

- **Server** — this PC writes trend logs and device data to the shared database. Typically the main PC that hosts the SQL Server instance. Data is written to both local SQLite and the server database for redundancy.
- **Client** — this PC reads trend logs and device data from the shared database. It continues to write its own local data to SQLite for offline use.

Select **Server** for the PC hosting the SQL Server instance. Select **Client** for all other PCs.

Under **Backend Type**, select **Microsoft SQL Server**.


Step 5 — Server Database enabled, Server role selected, Microsoft SQL Server backend

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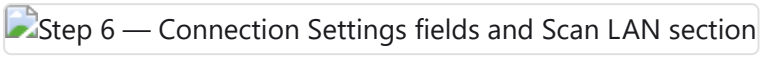
Step 6: Enter Connection Settings

Fill in the **Connection Settings — Microsoft SQL Server** fields:

Field	Description	Example
Host / IP	IP address or hostname of the SQL Server machine	`192.168.1.100`
Port	TCP port SQL Server is listening on	`1433`
Instance Name	SQL Server instance name	`SQLEXPRESS`
Database Name	Database to use (will be created by Init Schema)	`T3000`
Username	SQL login username	`sa`
Password	SQL login password	*(your password)*

Optionally use the **Connection URL** field to override all fields above with a single URL in the format `mssql://user:pass@host:port/dbname`.

To discover SQL Server instances automatically, click **Scan LAN**. The scanner probes ports 1432, 1433, and 1434 on your local subnet and returns all instances found. Click **Use** next to any result to fill in the connection fields automatically.


Step 6 — Connection Settings fields and Scan LAN section

Step 7: Use Scan LAN Results and Save

After scanning, the results show all discovered SQL Server instances. Click **Use** next to the instance you want to connect to. The connection fields are filled in automatically.

Once the fields are set, click **Save Configuration**. A success banner confirms:

> **Success** Server Database server mode + Microsoft SQL Server settings saved. Restart T3000 to apply.

> **Note:** If SQL Server Browser service is not running on the target machine, instances may appear as *TCP port check only* without an instance name. In this case, enter the instance name manually.


Step 7 — Scan results with 192.168.1.22:1432 selected, save success banner

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Step 8: Test Connection

Click **Test Connection** to verify the credentials and network connectivity.

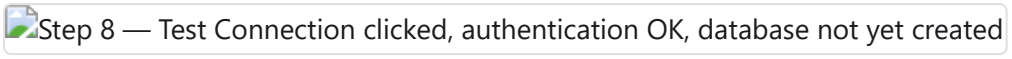
If the database does not exist yet, a notice appears:

> **Notice** Connected to SQL Server — authentication OK. Database 'T3000' does not exist yet. Click 'Init Schema' to create it.

This is expected for a first-time setup. Authentication succeeded — the database just needs to be initialized.

If connection fails, check:

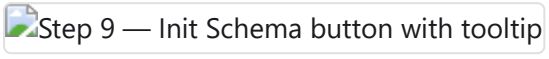
- SQL Server service is running
- TCP/IP is enabled in SQL Server Configuration Manager
- Windows Firewall allows the configured port (1433 or 1432)
- Username and password are correct



Step 9: Initialize the Schema

Click **Init Schema** to create all required T3000 tables (inputs, outputs, variables, trend logs, etc.) on the server database.

> Init Schema is safe to run multiple times. It will not delete existing data.



Step 10: Schema Initialization Complete

A success banner confirms:

> **Success** Schema initialized — 179 statements executed.

All T3000 tables have been created on the SQL Server database. The server is now ready to receive data.



Step 11: Restart T3000 to Activate

Return to the Dashboard. The **Shared DB** card now shows:

> *Shared DB (Server) saved — restart T3000 to activate.* **Database Configuration**

Close and reopen T3000 to apply the new configuration.


Step 11 — Dashboard showing Shared DB saved, restart required

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Step 12: Verify — Shared DB Active After Restart

After restarting T3000, the Dashboard shows **Shared DB** as **Active** with badges:

- **Server** — this PC is the server role
- **SQL Server** — connected to Microsoft SQL Server
- **Connected** — connection is live

The **Network Overview** section shows:

- **Server (This PC)** — Online, with the machine hostname and IP address
- *No client PCs connected yet. Clients appear once they send heartbeats.* — this will populate as Client PCs connect

T3000 Field Devices shows all devices in the building.


Step 12 — Dashboard after restart, Shared DB Active, Server Online, Network Overview

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Step 13: Confirm Sync & Database Health

Scroll down on the Dashboard to see:

System Overview shows:

- **Shared DB** — Connected
- **SQL Server · Server** — active backend and role
- **DB Size** — the current size of the SQL Server database (e.g. 16.0 MB at `D:\Program Files\Microsoft\...`)

Sync & Database Health shows:

- **Connected** badge, SQL Server, Direct write mode
- **Center DB Target** — `SQL Server · 192.168.x.x / T3000`
- Last Sync, Devices Today, Records Today — populate as sync cycles run

Trend Logs — Last 24 Hours shows trend data from all tracked points once devices begin polling:

- **Tracked** — number of tracked data points
- **Sampling Active** — points actively recording in the last 2 hours

- **Records (24h)** — total records written in the last 24 hours
- **Last Sample** — timestamp of the most recent data point


Step 13 — Dashboard System Overview Connected, Sync & Database Health, Trend Logs data

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Troubleshooting

If you run into issues, first check that SQL Server Express is correctly installed and configured — most connection problems start there. See the **SQL Server Express Setup guide** for the full installation and configuration walkthrough, including how to:

- Enable TCP/IP in SQL Server Configuration Manager
- Set a static port (1433 or 1432)
- Create a SQL login with the right permissions
- Open the correct port in Windows Firewall
- Start the SQL Server Browser service

Symptom	Likely Cause	Fix
Connection timeout	SQL Server service stopped or unreachable	Start the SQL Server service; verify network connectivity and firewall rules
Login failed	Wrong username or password	Check SQL login credentials; confirm **SQL Server Authentication** is enabled (not Windows-only)
Cannot connect remotely	Windows Firewall blocking the port	Allow TCP 1432 or 1433 inbound — see SQL Server Express Setup
Database does not exist	First-time setup, database not yet created	Run **Init Schema** after Test Connection shows "authentication OK"
Schema init error	Insufficient permissions on the SQL login	Ensure the SQL login has `CREATE DATABASE` and `CREATE TABLE` rights
Scan LAN finds no instances	SQL Server Browser service not running	Start the **SQL Server Browser** service on the target machine
Instance shows "TCP port check only"	SQL Browser not running, instance name unknown	Start SQL Server Browser, or enter the instance name manually in the connection fields
Dashboard still shows Standalone after save	T3000 not restarted after saving	Close and reopen T3000 to apply the new configuration
T3000 feature missing or not working	Running an older version of T3000	Click **Help — Check for Updates** and install the latest version

Summary

The following table shows the complete setup flow at a glance:

Step	Action	Result
1	Open T3000, click **Help — Check for Updates**	T3000 is on the latest version
2	Open Dashboard in browser (`localhost:9103` or `<this-PC-IP>:9103`)	Dashboard visible
3	Click **Shared DB** card — **Connect to Shared DB**	Opens Database Configuration page
4	Review Database Configuration page	Currently in Standalone mode
5	Enable **Server Database**, select **Server** role, select **Microsoft SQL Server**	Server DB enabled
6	Fill in Host / IP, Port, Instance Name, Database, Username, Password	Connection fields ready
7	Click **Scan LAN** — click **Use** next to target instance — click **Save Configuration**	Configuration saved
8	Click **Test Connection**	Authentication OK (database may not exist yet — that's normal)
9	Click **Init Schema**	Schema creation starts
10	Wait for success banner: *179 statements executed*	All T3000 tables created on SQL Server
11	Return to Dashboard, see restart prompt — close and reopen T3000	T3000 restarts with new config
12	Dashboard shows **Shared DB · Active · Connected**	Server is live and receiving data
13	Check **Sync & Database Health** and **Trend Logs** sections	Data flowing into SQL Server

> For Client PC setup, repeat Steps 1-13 on each client PC, selecting **Client** role in Step 5. No Init Schema is needed on client PCs.